

Security Alerting and Notification Implementation Artefact

PrimeCore Minerals Secure Multi Office Network

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Purpose and scope

This document records the HQ security alerting extension from a controlled pfSense block event to a received Grafana email and subsequent resolved notification. It supplies configuration details, defects, remediation and validation evidence for the implemented parts of SEC-01 and SEC-03. It also records storage remediation and verification of the central logging dependency.

Jared Williams documents the original HQ monitoring foundation in his HQ Network Monitoring and Logging Implementation artefact, including SPAN, rsyslog, Grafana, Loki and Alloy. My work builds on that foundation through storage remediation, validation, security alerting and email notification. Jared’s implementation is credited to him; his reported SPAN and access tests are not presented as tests I performed.

The scope is the demonstrated HQ test path. It excludes claims of completed incident acknowledgement, escalation, all management access paths, comprehensive retention, or authentication and shared-file auditing controls.

Environment and dependencies

Component	Role and observed configuration
HQ-MONITOR	Ubuntu 26.04; 192.168.90.60 on SERVER_VLAN 90; Grafana, Loki, Alloy and rsyslog
HQ-EMP-LAPTOP	Test source 192.168.11.100 on EMPLOYEE_WIFI_VLAN 11
Management gateway	Test destination 192.168.99.1 on MANAGEMENT_VLAN 99
pfSense R06	Logged employee WiFi to management block; tracker 1787829015
Collection	UDP 514 to rsyslog; /var/log/pfsense.log to Alloy, Loki and Grafana
Email	Gmail submission over TCP 587 with STARTTLS; PrimeCore IT Admin contact point

The alert consumes the central firewall log path. SPAN packet capture is a separate capability documented by Jared and is not a dependency of this alert query.

Infrastructure preparation and log verification

I expanded the HQ-MONITOR root logical volume and filesystem from approximately 14 GiB to 24 GiB using available volume-group space. The verification showed usage reduced from the previously recorded 92% to 54%, with approximately 11 GiB available and 4 GiB left in the volume group. This removed the immediate storage constraint without deleting logs.

I clarified the lab node identities and hostnames as HQ-MONITOR and HQ-EMP-LAPTOP. Address and interface checks established that HQ-MONITOR is 192.168.90.60 on SERVER_VLAN, the employee WiFi test laptop is

192.168.11.100, and the management gateway is 192.168.99.1. This corrected the earlier assumption that the monitoring server occupied the management VLAN.

I verified the existing collection path: pfSense records arrive through rsyslog on UDP 514, are written to /var/log/pfsense.log, read by Alloy and sent to Loki. Grafana successfully retrieves records using the job=pfsense and site=brisbane-hq labels. I saved the PrimeCore Security Monitoring dashboard with its HQ Firewall Logs panel.

I reviewed the logged R06 employee WiFi to management block rule and correlated its tracker 1787829015 with controlled test traffic. Four ping requests from 192.168.11.100 to 192.168.99.1 received no replies and produced four matching blocked records in Grafana. This verifies the tested ICMP path to the firewall's management address; it does not demonstrate every management destination or protocol.

SMTP configuration

The active /etc/grafana/grafana.ini SMTP section used enabled=true, host=smtp.gmail.com:587, user and from_address=primecoreitadmin@gmail.com, from_name=PrimeCore Security Monitoring, skip_verify=false and startTLS_policy=MandatoryStartTLS. The password was entered privately and is deliberately omitted. File ownership was set to root:grafana, mode 640, followed by a service restart and active-state check.

The first independently tested Google app password authenticated successfully. This does not establish that the later named app password was invalid. Successful Grafana test submission and mailbox receipt provide the application-level outcome.

Defects and remediation

Issue observed	Investigation and change	Verified outcome and limits
Low monitoring-server free space	Inspected disk and LVM layout; extended the root filesystem by 10 GiB	Root filesystem 24 GiB and 54% used
Grafana SMTP not configured	Inspected the active grafana.ini file; found SMTP values still commented with semicolons; enabled the settings	Subsequent attempts progressed to network and authentication errors; eventual mail delivery succeeded
Configuration access and restart	Set grafana.ini ownership to root:grafana and permissions to 640; restarted Grafana	Service reported active
SMTP TCP 587 timeout	Tested connectivity independently and reviewed SERVER_VLAN rules; added a logged allowance from HQ-MONITOR for TCP 587	Initial TCP connection succeeded
Recurring timeout as Gmail destination changed	Compared attempted destination 172.217.194.108 with alias table containing 172.217.194.109; a fixed-address retest then succeeded, but another destination later timed out	Static address additions did not provide a complete solution; final firewall rule capture remains outstanding
Gmail SMTP 534 5.7.9 WebLoginRequired	Isolated authentication with a direct STARTTLS test and privately entered app password	The first Google app password authenticated successfully; using the verified credential in Grafana led to test email delivery
LogQL parse error	Replaced the query with a corrected expression and source-IP parsing; verified it with controlled events	Counts returned for 192.168.11.100 and the alert later fired
Incorrect alert threshold and no-data settings	Changed threshold from above 0 to above 9, standardised severity label and set no-data handling to OK	Normal at 4 and 9 events; firing at 19; subsequent NoData resolution

For the repeated Gmail destination issue, the proposed lab remediation was a destination-any TCP 587 rule restricted to source 192.168.90.60. This would broaden allowed SMTP destinations for that host. Later successful email tests prove delivery for those attempts, but do not independently establish the final rule contents. The final rule must be captured and reviewed; a stable approved relay would permit a tighter destination policy.

The SMTP configuration uses smtp.gmail.com:587, mandatory STARTTLS and certificate verification. No app password is included in this report or its evidence.

Detection query

Query A is an instant query. The expression below filters the R06 tracker and IPv4 inbound block records, extracts source and destination IPv4 addresses, excludes parser errors and empty sources, and counts each source over five minutes.

```
sum by (src_ip) (
  count_over_time(
    {job="pfsense", site="brisbane-hq"}
    |= ",1787829015,"
    |= ",match,block,in,4,"
    | regexp `,(?P<src_ip>[0-9]{1,3}([.][0-9]{1,3}){3}),(?P<dst_ip>[0-9]{1,3}([.][0-9]{1,3}){3}),`
    | __error__=""
    | src_ip!=""
    [5m]
  )
)
```

The query groups by source IP, so the intended condition applies separately to each source. Multi-source separation has not yet been tested. The tracker is tied to the current firewall rule and must be reviewed if that rule is replaced.

Alert settings and validation

The saved rule is SEC-01 Repeated Employee WiFi Management Blocks, in the PrimeCore Security folder and Security Checks evaluation group. It counts matching R06 IPv4 inbound block records over five minutes, grouped by source IP. It evaluates every 30 seconds, uses a threshold above nine, and has no pending period or keep-firing period. Labels identify procedure SEC-01, medium severity and site brisbane-hq. Notifications use the PrimeCore IT Admin contact point.

Test	Observed result	Status
T01 controlled firewall block	Four unanswered pings and four correlated firewall records	Passed for tested path
T02 manual contact point	Grafana test succeeded and TestAlert email reached the inbox	Passed
T03 below threshold	Four records for source 192.168.11.100; Normal	Passed
T04 threshold comparison	Nine records remained Normal; ten additional pings produced 19 records and a true firing condition	Passed at 9 and 19; exact 10 boundary untested
T05 scheduled notification and recovery	Scheduled instance entered Alerting; rule-specific email showed A=19 and C=1; resolved email followed and instance became Normal NoData	Passed for demonstrated notification cycle

The scheduled instance showed firing at 17:14:40 in the browser and the firing email appeared at approximately 17:15. The instance later showed Normal NoData at 17:19:40 and a resolved email appeared at approximately 17:20. These observations are evidence of receipt, not an independently measured delivery service target.

Resolution used the configured NoData-to-OK policy, consistent with matching events leaving the rolling window. The resolved email's A=-1 and C=-1 are not negative event counts. Automatic resolution does not demonstrate human acknowledgement or incident closure, and no-data handling cannot establish collector health.

Administrator response and operational limits

An acknowledgement and response record has been drafted for the authorised test, with a proposed local incident register at deliverables/Security_Incident_Register.md. The actual acknowledgement time and completed investigation record have not yet been supplied. The draft response must not be reported as an executed incident-handling test.

Remaining work includes incorporating and reviewing Jared's setup documentation and Duane's completed network design; capturing the final SMTP firewall rule; testing the exact ten-event threshold and separation between sources; completing an actual acknowledgement record and escalation exercise; and validating logging health, retention, file auditing and authentication controls. Procedure approval and the remaining SEC-02 to SEC-05 acceptance tests must be tracked separately from document preparation.

Change control and security considerations

The work reused Jared's deployed logging foundation and verified each layer before proceeding to the next. The Grafana configuration was backed up using a non-overwriting copy command before SMTP work; backup existence should be checked before relying on it for restoration. Configuration secrets are not included in the artefact.

Successful network and email tests establish the particular attempts shown. The final egress policy still requires capture. NoData mapped to OK can resolve this rule when records disappear, so independent logging-health monitoring remains necessary before treating absence of security events as assurance of normal operation.

Sources and contribution record

Jared Williams, HQ Network Monitoring and Logging Implementation, supplied 13-page PDF. Its documented SPAN, rsyslog and Grafana stack form the baseline; Adam's screenshots document the subsequent changes and tests described here.

Evidence consists of screenshots supplied by Adam from the lab and recipient inbox. AI assisted with configuration guidance, troubleshooting, query correction and documentation. Adam executed the demonstrated lab actions. A completed response record and final policy evidence remain outstanding.

Figure 1 Storage resolution

The expanded root filesystem and remaining volume-group capacity verify the storage change.

```
labadmin@hq-monitor:~$ sudo lvextend -r -L +10G /dev/ubuntu-vg/ubuntu-lv
[sudo: authenticate] Password:
  File system ext4 found on ubuntu-vg/ubuntu-lv mounted at /.
  Size of logical volume ubuntu-vg/ubuntu-lv changed from <14.00 GiB (3583 extents) to <24.00 GiB (6143 extents).
  Extending file system ext4 to <24.00 GiB (25765609472 bytes) on ubuntu-vg/ubuntu-lv...
resize2fs /dev/ubuntu-vg/ubuntu-lv
resize2fs 1.47.2 (1-Jan-2025)
Filesystem at /dev/ubuntu-vg/ubuntu-lv is mounted on /; on-line resizing required
old_desc_blocks = 2, new_desc_blocks = 3
The filesystem on /dev/ubuntu-vg/ubuntu-lv is now 6290432 (4k) blocks long.

resize2fs done
  Extended file system ext4 on ubuntu-vg/ubuntu-lv.
  Logical volume ubuntu-vg/ubuntu-lv successfully resized.
labadmin@hq-monitor:~$ df -h /
Filesystem                                Size  Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv         24G   12G   11G   54% /
labadmin@hq-monitor:~$ sudo lvs
  LV      VG      Attr      LSize   Pool Origin Data%  Meta%   Move Log Cpy%Sync Convert
ubuntu-lv ubuntu-vg -wi-ao---- <24.00g
labadmin@hq-monitor:~$ sudo vgs
  VG      #PV #LV #SN Attr   VSize   VFree
ubuntu-vg    1  1   0 wz--n- <28.00g 4.00g
labadmin@hq-monitor:~$
```

Figure 2 Verified network interfaces

The interface map distinguishes SERVER_VLAN from MANAGEMENT_VLAN.

```
WAN (wan) -> vtnet0 -> v4/DHCP4: 192.168.1.159/24
LAN (lan) -> vtnet1 -> v4: 10.10.10.1/24
EMPLOYEE_VLAN (opt1) -> vtnet2.10 -> v4: 192.168.10.1/24
VISITOR_VLAN (opt2) -> vtnet2.20 -> v4: 192.168.20.1/24
EMPLOYEE_WIFI_VLAN (opt3) -> vtnet2.11 -> v4: 192.168.11.1/24
PRINTER_VLAN (opt4) -> vtnet2.30 -> v4: 192.168.30.1/24
VIDEO_VLAN (opt5) -> vtnet2.40 -> v4: 192.168.40.1/24
CCTV_VLAN (opt6) -> vtnet2.41 -> v4: 192.168.41.1/24
IOT_VLAN (opt7) -> vtnet2.50 -> v4: 192.168.50.1/24
SERVER_VLAN (opt8) -> vtnet2.90 -> v4: 192.168.90.1/24
BACKUP_VLAN (opt9) -> vtnet2.91 -> v4: 192.168.91.1/24
MANAGEMENT_VLAN (opt10) -> vtnet2.99 -> v4: 192.168.99.1/24

0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell

9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + Netgate pfSense Plus tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM

Enter an option: █
```

Figure 3 Existing collection configuration

Alloy reads the pfSense log file and forwards labelled events to Loki.

```
labadmin@hq-monitor:~$ sudo less /etc/alloy/config.alloy
loki.source.file "pfsense" {
  targets = [
    {
      __path__ = "/var/log/pfsense.log",
      job      = "pfsense",
      site     = "brisbane-hq",
    },
  ]

  forward_to = [loki.write.local.receiver]
}

loki.write "local" {
  endpoint {
    url = "http://127.0.0.1:3100/loki/api/v1/push"
  }
}
/etc/alloy/config.alloy (END)
```

Figure 4 Saved monitoring dashboard

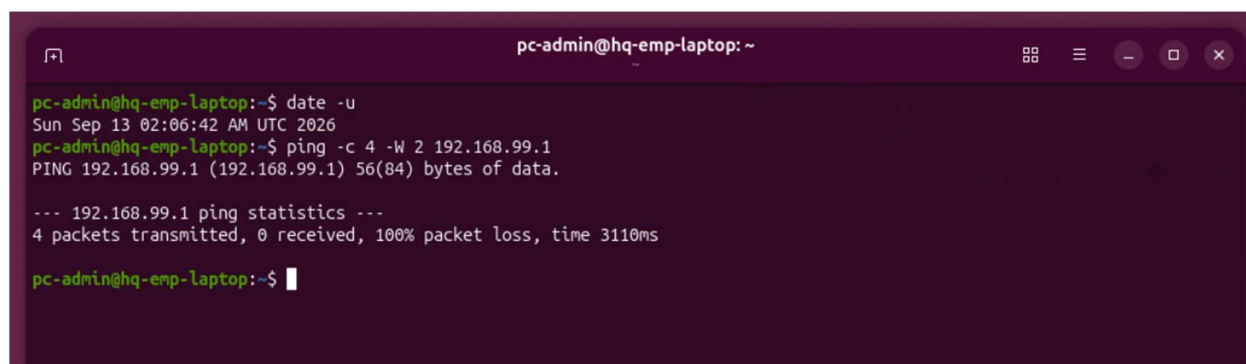
Firewall records are available in the saved dashboard.

The screenshot displays a web browser window with the URL `http://192.168.90.60:3090/edges/primecore-security-monitoring/FromNow-6H&is-new&timezone-browser`. The browser's address bar shows a VPN icon. The dashboard header includes the text "Dashboards > PrimeCore Security Monitoring" and a search bar. Below the header, there are navigation controls: "Last 6 hours", "Refresh", and a dropdown menu. The main content area is titled "HQ Firewall Logs" and displays a list of firewall log entries. Each entry is a single line of text representing a log record, including timestamps, log levels, and detailed firewall rule information.

```
HO Firewall Logs
[ UNK 2026-09-13T00:26:44+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,4433,0,none,17,udp,216,192.168.1.27,255.255.255.255,49154,6667,196
[ UNK 2026-09-13T00:26:43+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,37132,0,DF,17,udp,216,192.168.1.68,255.255.255.255,57247,6667,196
[ UNK 2026-09-13T00:26:43+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,65390,0,DF,17,udp,200,192.168.1.229,255.255.255.255,49597,6667,180
[ UNK 2026-09-13T00:26:43+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,14446,0,DF,17,udp,32,192.168.1.32,255.255.255.255,60865,10001,12
[ UNK 2026-09-13T00:26:43+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,61143,0,DF,17,udp,63,192.168.1.25,192.168.1.255,35003,15600,43
[ UNK 2026-09-13T00:26:42+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,128,57218,0,none,17,udp,72,192.168.1.237,192.168.1.255,57621,57621,52
[ UNK 2026-09-13T00:26:42+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,169,0,none,17,udp,216,192.168.1.28,255.255.255.255,49154,6667,196
[ UNK 2026-09-13T00:26:42+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,38694,0,none,17,udp,200,192.168.1.99,255.255.255.255,65060,6667,180
[ UNK 2026-09-13T00:26:41+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,0,0,DF,17,udp,248,192.168.1.221,255.255.255.255,54286,6667,228
[ UNK 2026-09-13T00:26:41+00:00 _gateway filterlog[57828]: 1.6.,1000000103,vtnet2_match,block,in,4,0x0.,64,0,0,none,17,udp,313,0.0.0.0,255.255.255.255,68,67,293
[ UNK 2026-09-13T00:26:41+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,63407,0,none,17,udp,216,192.168.1.26,255.255.255.255,49154,6667,196
[ UNK 2026-09-13T00:26:40+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,4432,0,none,17,udp,216,192.168.1.27,255.255.255.255,49154,6667,196
[ UNK 2026-09-13T00:26:39+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,37131,0,DF,17,udp,216,192.168.1.68,255.255.255.255,57247,6667,196
[ UNK 2026-09-13T00:26:39+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,65005,0,DF,17,udp,200,192.168.1.229,255.255.255.255,49597,6667,180
[ UNK 2026-09-13T00:26:37+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,60522,0,DF,17,udp,63,192.168.1.25,192.168.1.255,40692,15600,43
[ UNK 2026-09-13T00:26:37+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,168,0,none,17,udp,216,192.168.1.28,255.255.255.255,49154,6667,196
[ UNK 2026-09-13T00:26:37+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,38693,0,none,17,udp,200,192.168.1.99,255.255.255.255,65060,6667,180
[ UNK 2026-09-13T00:26:37+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,0,0,DF,17,udp,248,192.168.1.221,255.255.255.255,54286,6667,228
[ UNK 2026-09-13T00:26:36+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,63406,0,none,17,udp,216,192.168.1.26,255.255.255.255,49154,6667,196
[ UNK 2026-09-13T00:26:35+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,4431,0,none,17,udp,216,192.168.1.27,255.255.255.255,49154,6667,196
[ UNK 2026-09-13T00:26:34+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,37130,0,DF,17,udp,216,192.168.1.68,255.255.255.255,57247,6667,196
[ UNK 2026-09-13T00:26:33+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,64855,0,DF,17,udp,200,192.168.1.229,255.255.255.255,49597,6667,180
[ UNK 2026-09-13T00:26:33+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,14258,0,DF,17,udp,32,192.168.1.32,255.255.255.255,46329,10001,12
[ UNK 2026-09-13T00:26:32+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,167,0,none,17,udp,216,192.168.1.28,255.255.255.255,49154,6667,196
[ UNK 2026-09-13T00:26:32+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,38692,0,none,17,udp,200,192.168.1.99,255.255.255.255,65060,6667,180
[ UNK 2026-09-13T00:26:32+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,0,0,DF,17,udp,248,192.168.1.221,255.255.255.255,54286,6667,228
[ UNK 2026-09-13T00:26:31+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,59041,0,DF,17,udp,63,192.168.1.25,192.168.1.255,45620,15600,43
[ UNK 2026-09-13T00:26:31+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,59002,0,none,17,udp,57,192.168.1.254,255.255.255.255,9999,9999,37
[ UNK 2026-09-13T00:26:31+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,63405,0,none,17,udp,216,192.168.1.26,255.255.255.255,49154,6667,196
[ UNK 2026-09-13T00:26:30+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,255,4430,0,none,17,udp,216,192.168.1.27,255.255.255.255,49154,6667,196
[ UNK 2026-09-13T00:26:29+00:00 _gateway filterlog[57828]: 1.71.,12014,vtnet0_match,block,in,4,0x0.,64,37129,0,DF,17,udp,216,192.168.1.68,255.255.255.255,57247,6667,196
```

Figure 5 Controlled blocked traffic

The initial four ping requests receive no replies.

A terminal window titled 'pc-admin@hq-emp-laptop: ~' with standard window controls. The terminal shows a sequence of commands and their outputs. First, the user runs 'date -u', which returns 'Sun Sep 13 02:06:42 AM UTC 2026'. Then, the user runs 'ping -c 4 -W 2 192.168.99.1'. The output shows 'PING 192.168.99.1 (192.168.99.1) 56(84) bytes of data.' followed by a separator '--- 192.168.99.1 ping statistics ---'. The statistics show '4 packets transmitted, 0 received, 100% packet loss, time 3110ms'. Finally, the prompt 'pc-admin@hq-emp-laptop:~\$' is shown with a cursor.

```
pc-admin@hq-emp-laptop:~$ date -u
Sun Sep 13 02:06:42 AM UTC 2026
pc-admin@hq-emp-laptop:~$ ping -c 4 -W 2 192.168.99.1
PING 192.168.99.1 (192.168.99.1) 56(84) bytes of data.

--- 192.168.99.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3110ms

pc-admin@hq-emp-laptop:~$
```

Figure 6 Correlated firewall evidence
Four matching records support the controlled block test.

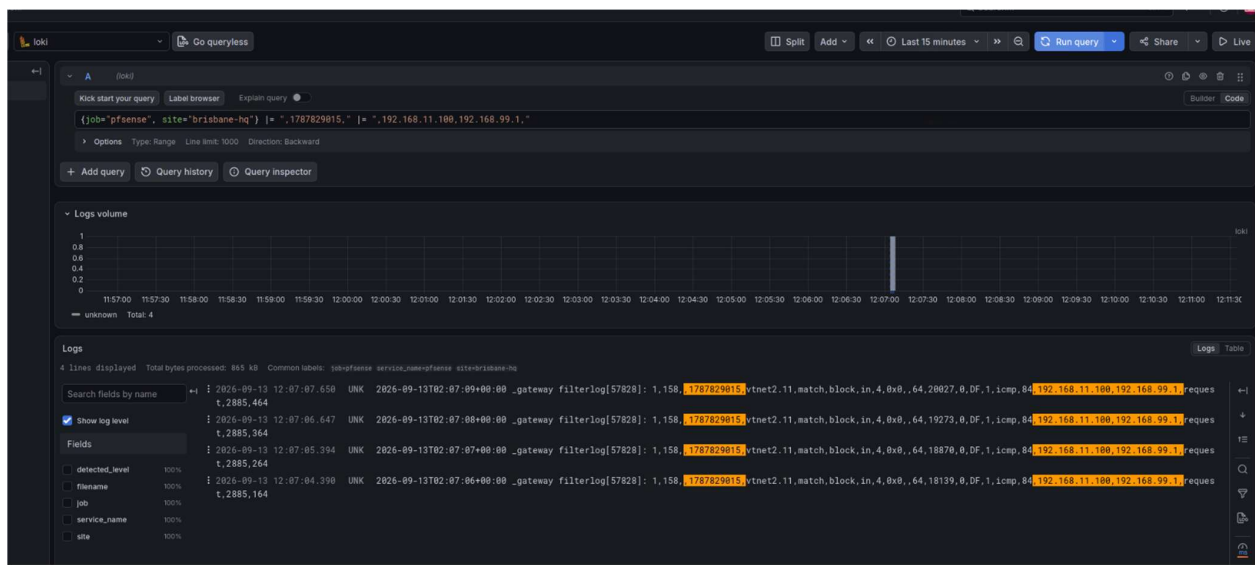


Figure 7 SMTP configuration error

Grafana reports that SMTP is not configured.

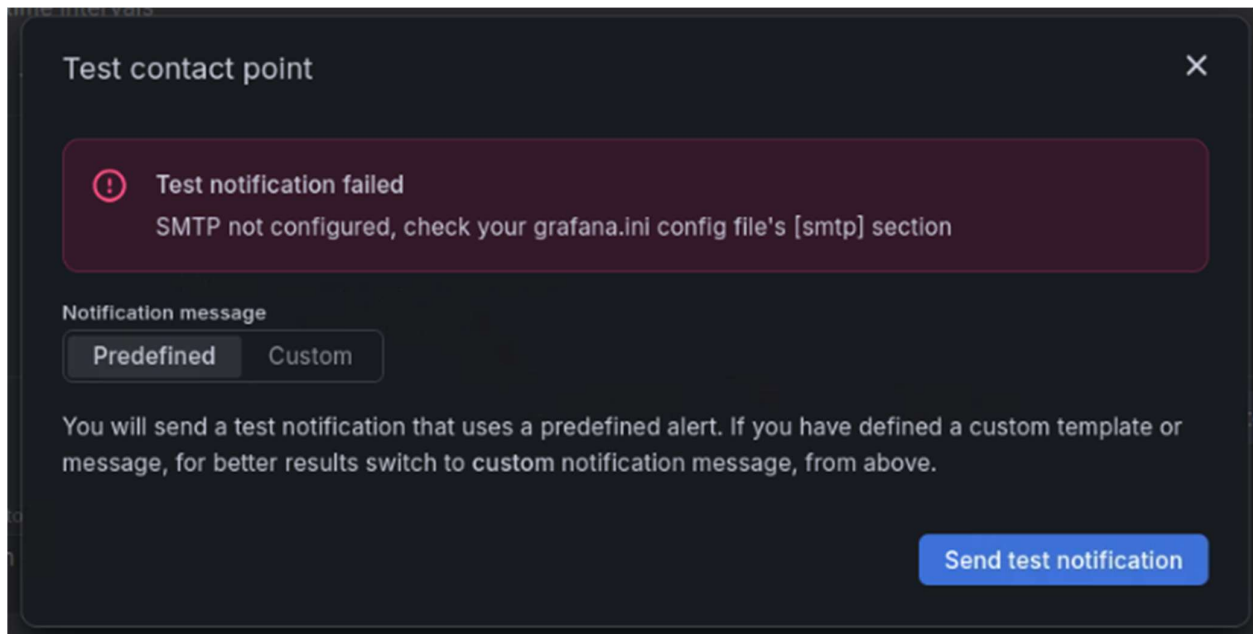


Figure 8 Commented SMTP settings

The visible semicolons explain why those configuration values were inactive.

```
labadmin@hq-monitor:~$ sudo grep -nE '^[[:space:]]*[:#]?[[:space:]]*\[smtp\]' /etc/grafana/grafana.ini
1155:[smtp]
labadmin@hq-monitor:~$ sudo sed -n '/^\[smtp\]/,/^\[/p' /etc/grafana/grafana.ini | grep -E '^[[:space:]]*[:#]?[[:space:]]*(enabl
ed|host|from_address|startTLS_policy)[[:space:]]*='
;enabled = true
;host = smtp.gmail.com:587
;from_address = primecoreitadmin@gmail.com
;startTLS_policy = MandatoryStartTLS
labadmin@hq-monitor:~$ sudo journalctl -u grafana-server -b --no-pager | grep 'Config loaded from' | tail -n 5
Sep 13 04:30:53 hq-monitor grafana[1528]: logger=settings t=2026-09-13T04:30:53.872014353Z level=info msg="Config loaded from" f
ile=/etc/grafana/grafana.ini
Sep 13 04:34:03 hq-monitor grafana[1730]: logger=settings t=2026-09-13T04:34:03.246307047Z level=info msg="Config loaded from" f
ile=/usr/share/grafana/conf/defaults.ini
Sep 13 04:34:03 hq-monitor grafana[1730]: logger=settings t=2026-09-13T04:34:03.246319689Z level=info msg="Config loaded from" f
ile=/etc/grafana/grafana.ini
Sep 13 04:38:18 hq-monitor grafana[1925]: logger=settings t=2026-09-13T04:38:18.505551384Z level=info msg="Config loaded from" f
ile=/usr/share/grafana/conf/defaults.ini
Sep 13 04:38:18 hq-monitor grafana[1925]: logger=settings t=2026-09-13T04:38:18.505569312Z level=info msg="Config loaded from" f
ile=/etc/grafana/grafana.ini
labadmin@hq-monitor:~$ _
```

Figure 9 Permissions and restart verification

Ownership and permissions were set and the restarted service reports active.

```
labadmin@hq-monitor:~$ sudo chown root:grafana /etc/grafana/grafana.ini
[sudo: authenticate] Password:
labadmin@hq-monitor:~$ sudo chmod 640 /etc/grafana/grafana.ini
labadmin@hq-monitor:~$ sudo systemctl restart grafana-server
labadmin@hq-monitor:~$ sudo systemctl is-active grafana-server
active
labadmin@hq-monitor:~$
```

Figure 10 SMTP transport failure

The contact-point test times out before authentication.

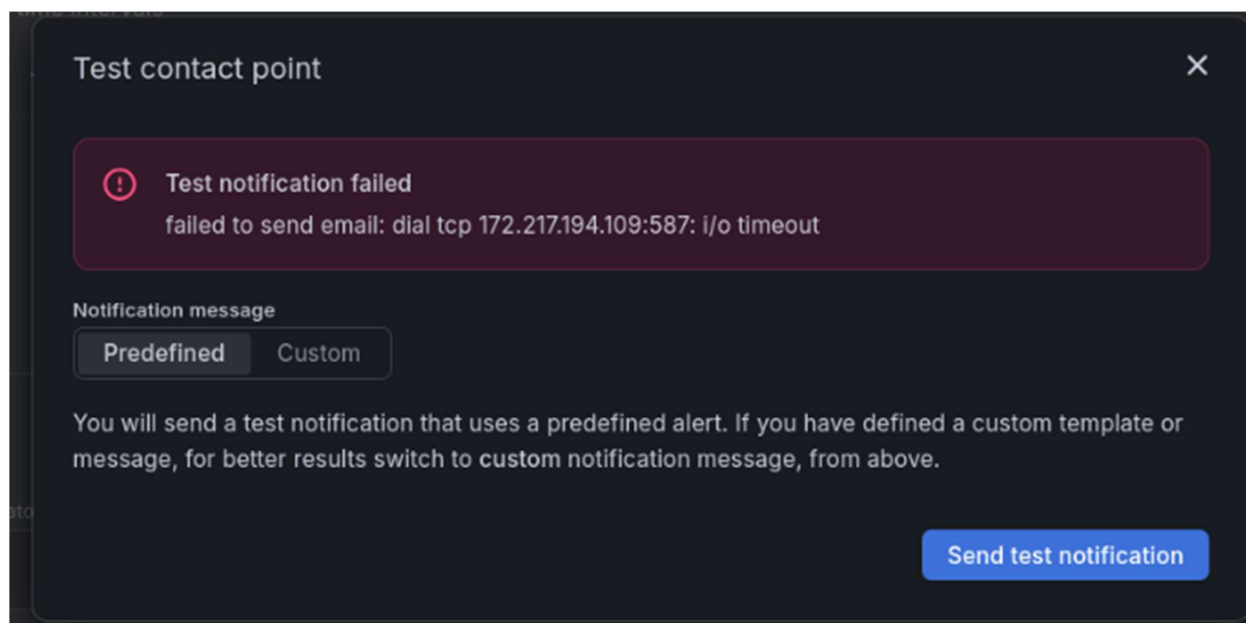


Figure 11 Initial SMTP firewall allowance

The initial logged rule permits HQ-MONITOR to the Gmail alias on TCP 587; this is not a capture of the final policy.

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

SERVER_VLAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

Address or Alias

192.168.90.60

/

Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination

Destination

☐ Invert match

Address or Alias

GMAIL_SMTP

/

Destination Port Range

SUBMISSION (587)

From

Custom

SUBMISSION (587)

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☒ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

Allow HQ-MONITOR Gmail SMTP

A description may be entered here for administrative reference. A maximum of 57 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

Rule Information

Tracking ID

1789275475

Created

9/13/26 04:57:55 by admin@192.168.99.71 (Local Database)

Updated

9/13/26 04:57:55 by admin@192.168.99.71 (Local Database)

Figure 12 Initial connectivity resolution

The independent TCP test succeeds after the initial rule change.

```
labadmin@hq-monitor:~$ date -u
Sun Sep 13 04:51:02 AM UTC 2026
labadmin@hq-monitor:~$ nc -vz -w 5 smtp.gmail.com 587
nc: connect to smtp.gmail.com (172.217.194.109) port 587 (tcp) timed out: Operation now in progress
nc: connect to smtp.gmail.com (2404:6800:4003:c05::6d) port 587 (tcp) failed: Network is unreachable
labadmin@hq-monitor:~$ nc -4 -vz -w 5 smtp.gmail.com 587
Connection to smtp.gmail.com (172.217.194.108) 587 port [tcp/submission] succeeded!
labadmin@hq-monitor:~$ _
```

Figure 13 Destination alias mismatch

The displayed alias includes 172.217.194.109 while the failed destination was 172.217.194.108.

pfSense +

System ▾

Interfaces ▾

Firewall ▾

Services ▾

VPN ▾

Status ▾

Diagnostics ▾

Help ▾

Diagnostics / Tables

Table to Display

Table

GMAIL_SMTP ▾

Empty Table

Select a user-defined alias name or system table name to view its contents.

Aliases become Tables when loaded into the active firewall ruleset. The contents displayed on this page reflect the current addresses inside tables used by the firewall.

i

Date of last update of table is unknown. 2 records.

GMAIL_SMTP Table

IP Address

172.217.194.109

2404:6800:4003:c04::6d

Figure 14 Fixed destination retest

Connection to the previously failing address succeeds for this attempt.

```
labadmin@hq-monitor:~$ nc -4 -vz -w 5 172.217.194.108 587
Connection to 172.217.194.108 587 port [tcp/submission] succeeded!
labadmin@hq-monitor:~$ _
```

Figure 15 Recurring destination failure

A new Gmail destination times out, exposing the limit of the temporary address remedy.

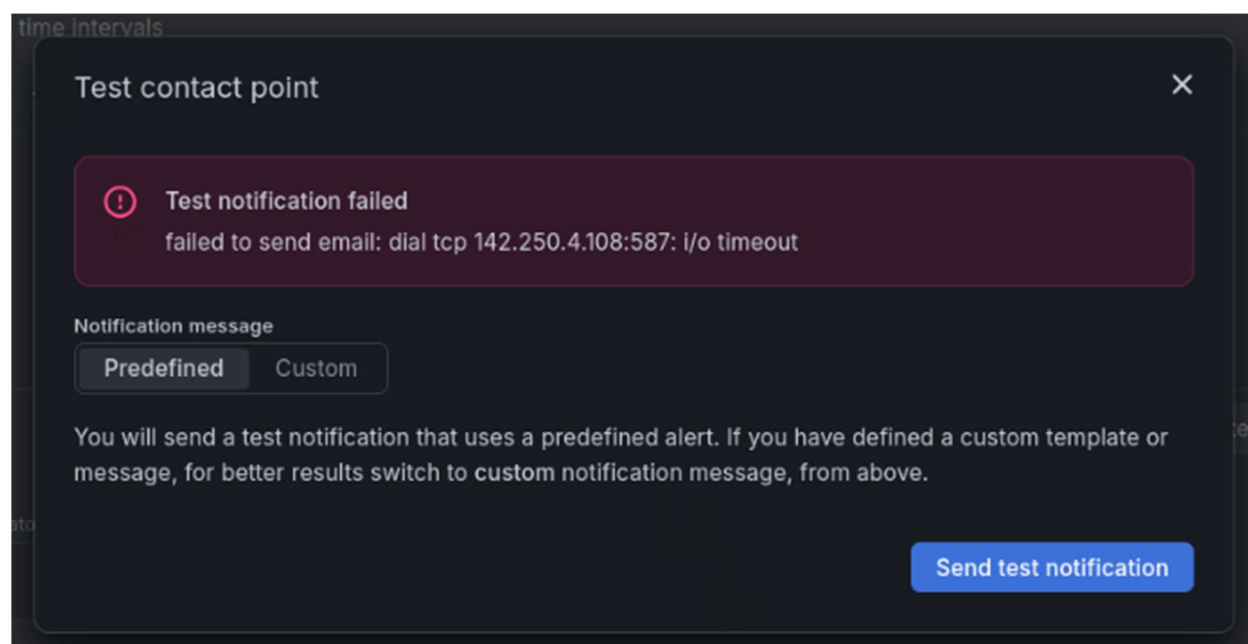


Figure 16 Gmail authentication error

The 534 response confirms an authentication-stage failure rather than a TCP timeout.

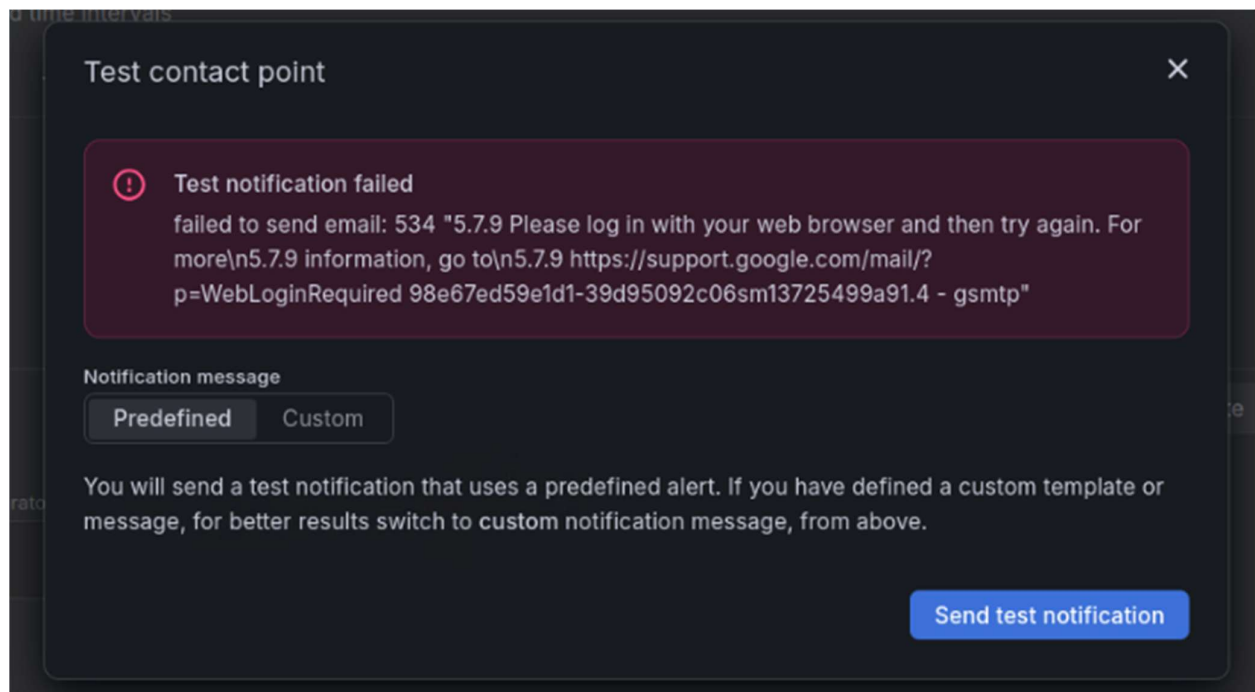


Figure 17 Independent authentication resolution

The direct test confirms successful authentication without displaying the password.

```
labadmin@hq-monitor:~$ python3 -c 'import smtplib,ssl,getpass; p=getpass.getpass("Gmail app password: ").replace("'", ""); s=smtplib.SMTP("smtp.gmail.com",587,timeout=15); s.ehlo(); s.starttls(context=ssl.create_default_context()); s.ehlo(); s.login("primeco-reitadmin@gmail.com",p); print("AUTHENTICATION SUCCEEDED"); s.quit()'  
Gmail app password:  
AUTHENTICATION SUCCEEDED  
labadmin@hq-monitor:~$ _
```

Figure 18 Successful Grafana test

Grafana confirms test submission succeeded.

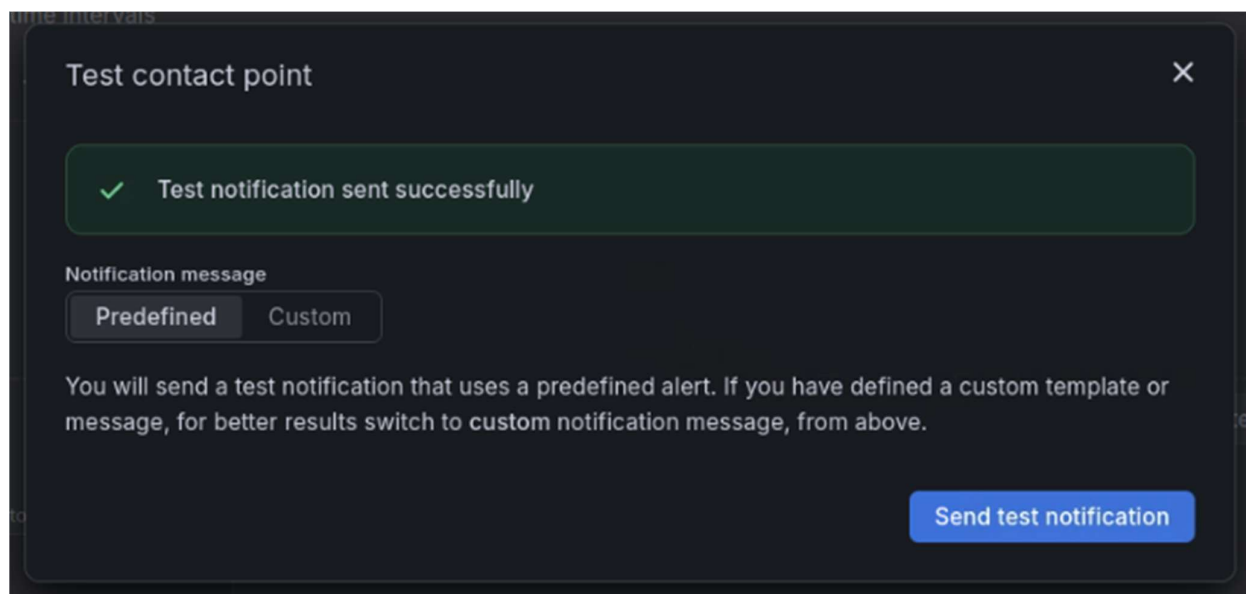


Figure 19 Manual test email receipt

The inbox confirms receipt of the manual notification.

[FIRING:1] TestAlert Grafana

Inbox x

PrimeCore Security Monitoring <primecoreitadmin@gmail.com>

to me



Grouped by

alertname=TestAlert instance=Grafana

1 firing instances

Firing	TestAlert
Summary Notification test Labels alertname instance TestAlert Grafana	
Silence	
Observed 0s before this notification was delivered, at 2026-09-13 06:32:31.703335132 +0000 UTC m=+39.470054745	

Figure 20 Alert query error

The initial query fails parsing and requires correction.



Figure 21 Initial alert settings

The saved threshold was above zero before correction.

Alerting > PrimeCore Security > Security Checks > SEC-01 Repeated Employee WiFi Management Blocks

Q Search...

ctrl+k

+

-

🔔

< SEC-01 Repeated Employee WiFi Management Blocks Normal

Notifications are delivered to PrimeCore IT Admin

Evaluation interval: Every 30s

Labels: Severity: medium procedure: SEC-01 site: brisbane-hq Edit More

Query and conditions

Instances

History

Versions

A to now

View in Explore

```
sum by (src_ip) (count_over_time({job="pfSense", site="brisbane-hq"} |= ",1787829015," |= ",match_block_in_4," | regexp ",(IP+src_ip[0-9]{1,3}([0-9]{1,3}){3}), (IP+dst_ip[0-9]{1,3}([0-9]{1,3}){3}),
| _error_" | src_ip" [5m]))
```

Table

No data

C Threshold Alert condition

Input

A Is above 0

No data

Rule type

Grafana-managed alert rule

Rule Identifier

afy3vzsrk4g8e

Last updated by

admin

Last updated at

2026-09-13 16:58:17 (a few seconds ago)

Evaluation

Last evaluated

-

Last evaluation duration

0s

Pending period

0s

Keep firing for

0s

Notification configuration

Contact point

PrimeCore IT Admin

Alert state

Alert state if no data or all values are null

NoData

Alert state if execution error or timeout

Error

Annotations

No annotations

Adam Donovan | Security Alerting | 25

Figure 22 Corrected alert settings

The corrected threshold is above nine and no-data handling is OK.

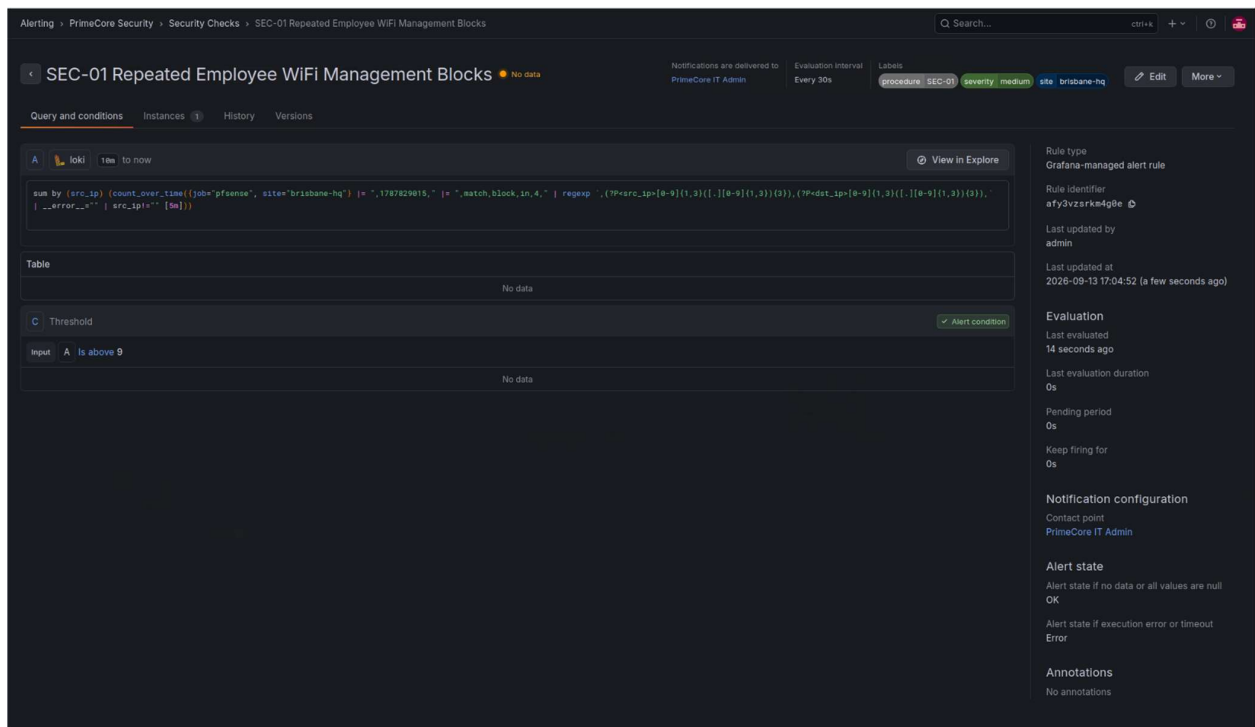


Figure 23 Query resolution and low count test

The corrected query returns four records for the expected source and remains Normal.

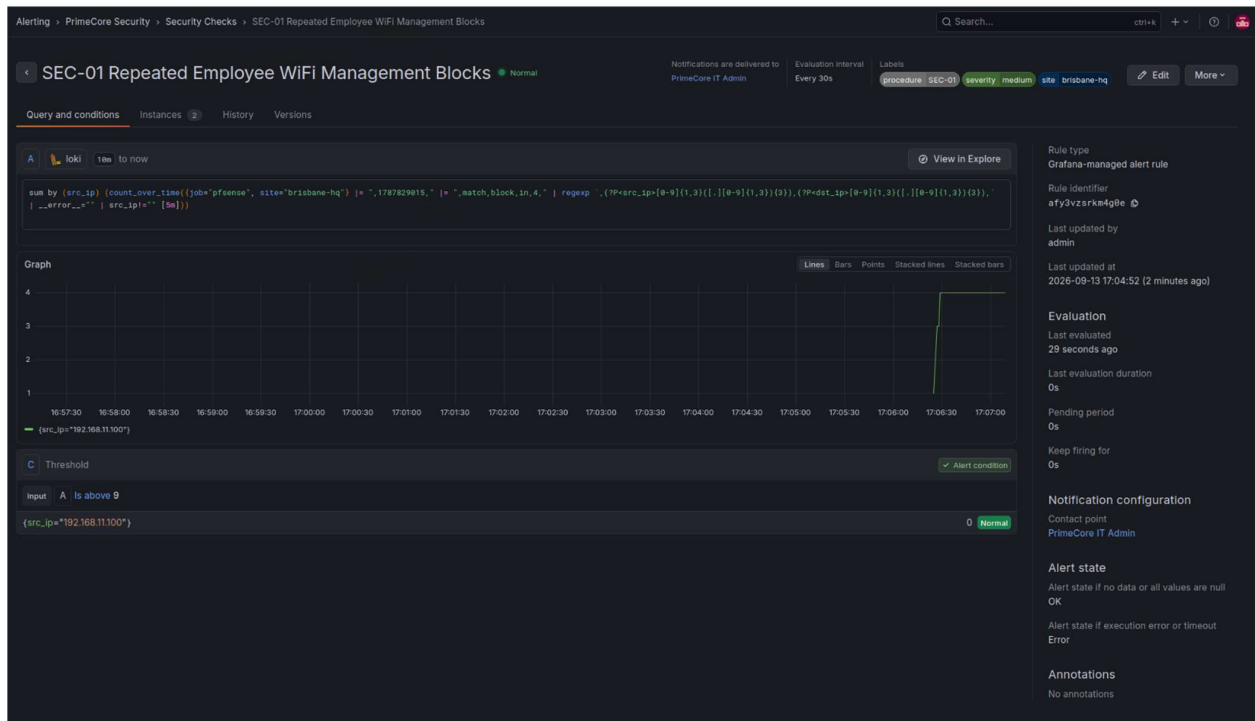


Figure 24 Nine event comparison
Nine matching events remain below the firing condition.

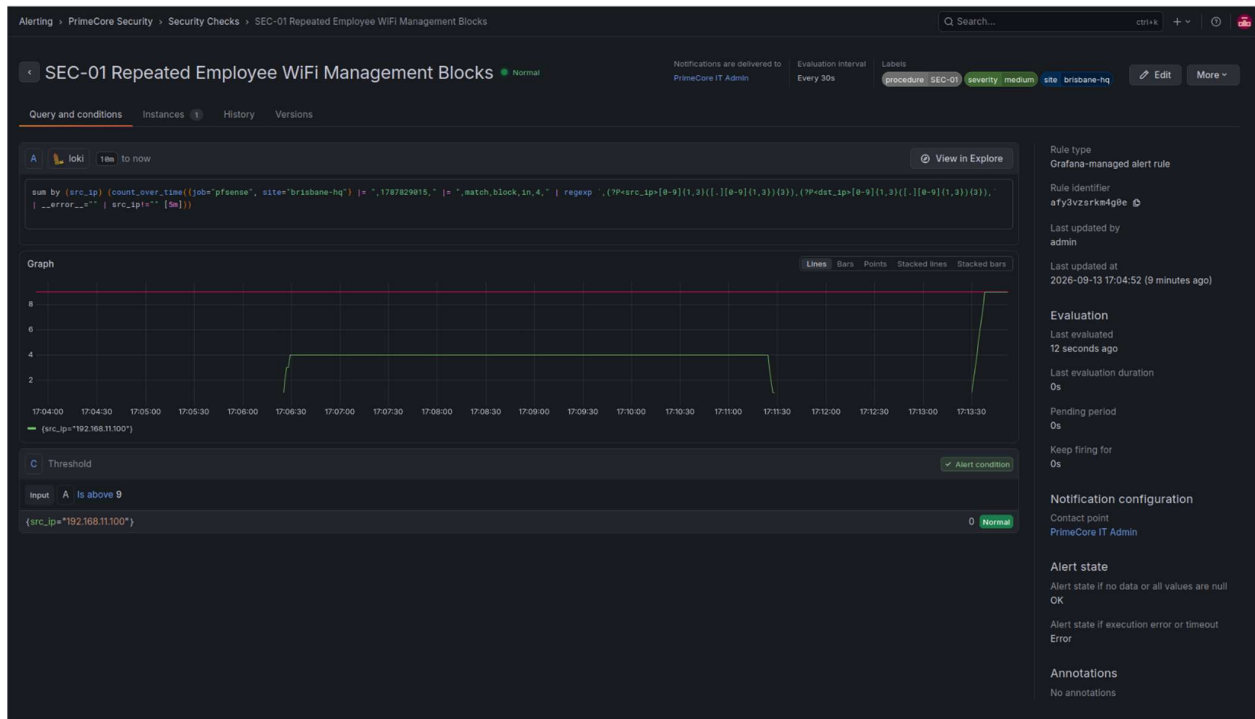
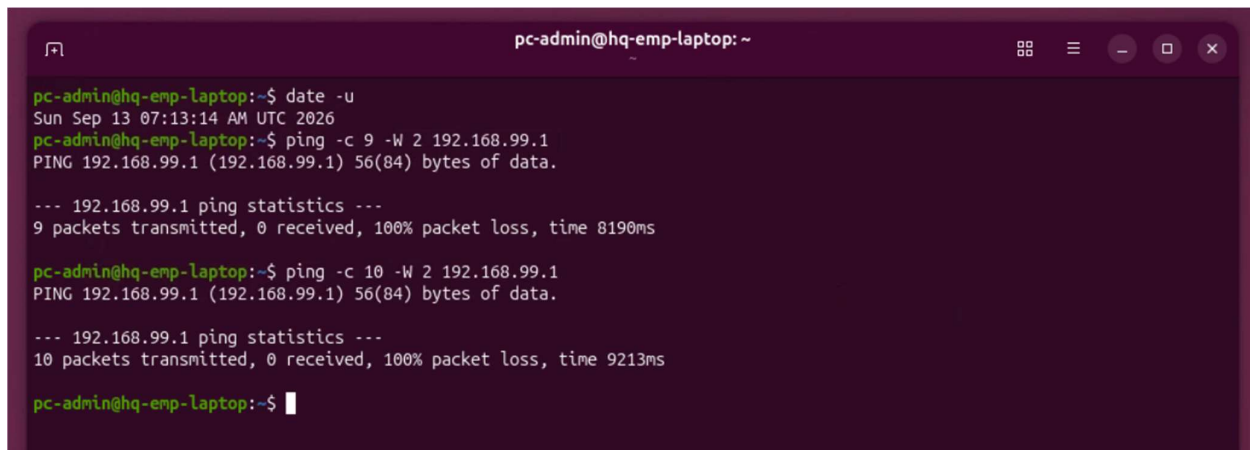


Figure 25 Actual threshold test traffic

Nine requests followed by ten additional requests explain the observed total of 19.

A terminal window titled 'pc-admin@hq-emp-laptop: ~' with standard window controls. It shows two ping commands and their statistics. The first command is 'ping -c 9 -W 2 192.168.99.1', resulting in 9 packets transmitted, 0 received, and 100% packet loss. The second command is 'ping -c 10 -W 2 192.168.99.1', resulting in 10 packets transmitted, 0 received, and 100% packet loss. The prompt is at the end of the second command.

```
pc-admin@hq-emp-laptop:~$ date -u
Sun Sep 13 07:13:14 AM UTC 2026
pc-admin@hq-emp-laptop:~$ ping -c 9 -W 2 192.168.99.1
PING 192.168.99.1 (192.168.99.1) 56(84) bytes of data.

--- 192.168.99.1 ping statistics ---
9 packets transmitted, 0 received, 100% packet loss, time 8190ms

pc-admin@hq-emp-laptop:~$ ping -c 10 -W 2 192.168.99.1
PING 192.168.99.1 (192.168.99.1) 56(84) bytes of data.

--- 192.168.99.1 ping statistics ---
10 packets transmitted, 0 received, 100% packet loss, time 9213ms

pc-admin@hq-emp-laptop:~$
```

Figure 26 Scheduled alert firing
The source-specific instance enters Alerting.

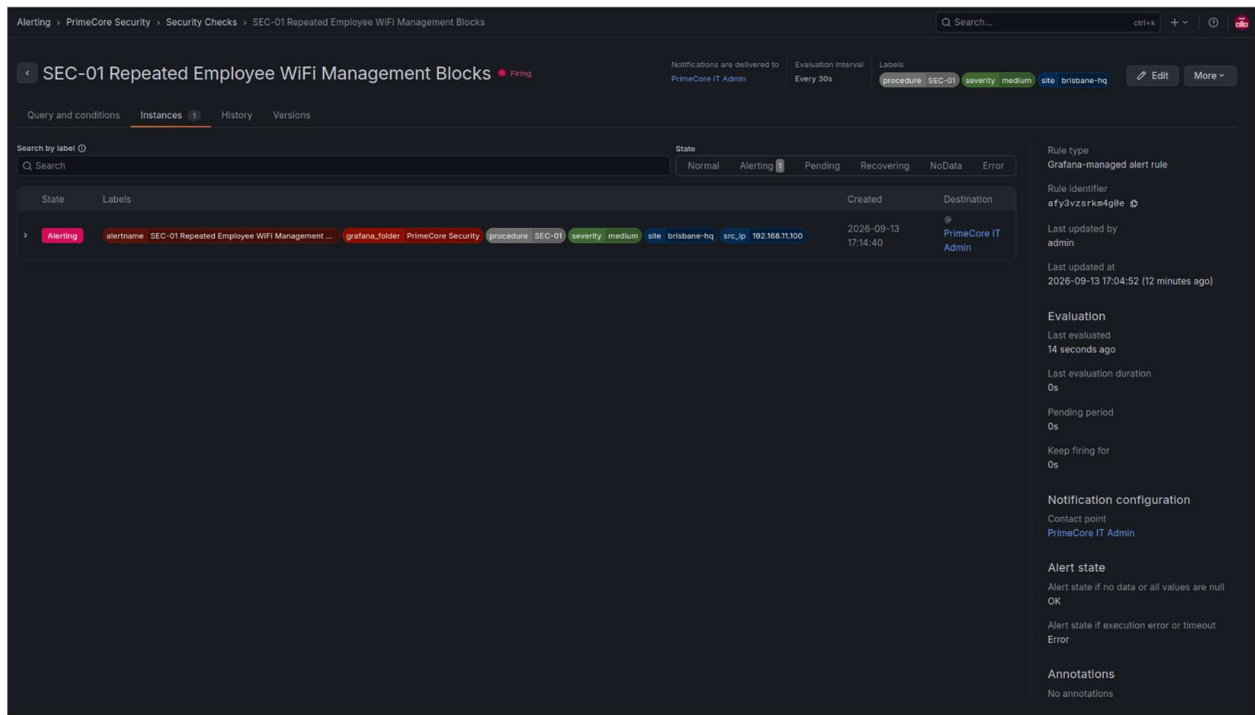


Figure 27 Rule triggered email receipt

The received notification reports A=19 and C=1.

[FIRING:1] SEC-01 Repeated Employee WiFi Management Blocks PrimeCore Security (SEC-01 medium brisbane-hq 192.168.11.100)

to me

17:15 (2 minutes ago)

PrimeCore Security Monitoring

<primecoreitadmin@gmail.com>

to me

PrimeCore Security

SEC-01 Repeated Employee WiFi Management Blocks

1 firing instances

Firing

SEC-01 Repeated Employee WiFi Management Blocks

View alert

Values

A=19 C=1

Labels

alertrnameSEC-01 Repeated Employee WiFi Management Blocks

grafana_folderPrimeCore Security

procedureSEC-01

severitymedium

sitebrisbane-hq

src_ip192.168.11.100

Silence

Observed 30s before this notification was delivered, at 2026-09-13 07:14:40 +0000 UTC

© 2026 Grafana Labs. Sent by Grafana v13.2.1.

Figure 28 Resolved email receipt

The received resolution identifies NoData as the state reason.

[RESOLVED] SEC-01 Repeated Employee WiFi Management Blocks PrimeCore Security (SEC-01 medium brisbane-hq 192.168.11.100)

Inbox

PrimeCore Security Monitoring

<primecoreitadmin@gmail.com>

to me

17:20 (1 minute ago)

☆

😊

↶

⋮

Grafana

PrimeCore Security > SEC-01 Repeated Employee WiFi Management Blocks

1 resolved instances

Resolved

SEC-01 Repeated Employee WiFi Management Blocks

View alert

Values

A=-1 C=-1

Labels

alernameSEC-01 Repeated Employee WiFi Management Blocks

grafana_folderPrimeCore Security

procedureSEC-01

severitymedium

sitebrisbane-hq

src_ip192.168.11.100

Annotations

datasource_uiddfxupb342ireoc

grafana_state_reasonNoData

ref_idA

Silence

Observed 5m30s before this notification was delivered, at 2026-09-13 07:14:40 +0000 UTC

© 2026 Grafana Labs. Sent by [Grafana v13.2.1](#).

Figure 29 Final recovery state

The instance returns to Normal NoData under the configured policy.

